

ASTR 210 Fall 2025 — Homework set 12
50 points plus optional extra credit
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Solutions need to show all important intermediate calculations, diagrams, and explanations in addition to the final answer to receive full credit. Please check the legibility of homework solutions uploaded to the class website as we cannot grade or give credit for illegible or unreadable work. Units do not need to be specified if an expression or equation is requested; for numerical calculations please use the specified units for the answer, or if not specified, SI or cgs units. Please use Table A.1 for the value of physical constants and Table A.2 for the value of astronomical constants, unless otherwise specified in the problem. Please use the correct number of significant figures.

12.1 Eddington luminosity and massive stars

High-mass stars are sufficiently luminous that radiation pressure can be an important component of their internal support against gravity. Is the Eddington limit also relevant for stars?

- (a) Answer this question by using the Eddington relations and the mass-luminosity relation for high-mass stars to determine the maximum mass of a star that is stable against disruption by radiation pressure. [7 pts]
- (b) Comment on how this limit compares to the most massive known stars. What are the most massive stars? There is some information in our text and some, of course, online. *You must provide a citation of some sort to get any credit for this part.* [3 pts]

12.2 Accretion disk temperatures

Your text works through the physics relevant to predicting the temperatures of the accretion disks around black holes.

- (a) Do the missing algebra steps to show how equation 21.21 can be combined with equation 21.22 to produce equation 21.23. [3 pts]
- (b) Now make a couple of helpful assumptions. First, assume that the black hole is accreting at its Eddington rate (so that its luminosity is equal to the Eddington luminosity). For simplicity, you can take the efficiency parameter $\eta \sim 0.1$. Then assume that most of the radiation comes from near the black hole's *innermost stable circular orbit*, which is at $r = 3r_{Sch} = 6GM/c^2$ for a non-rotating black hole. Use that information to show that the temperature of the accretion disk should scale with the black hole's mass as $T \propto M^{-1/4}$. [5 pts]
- (c) What temperatures should we expect for the accretion disks around a $4 M_{\odot}$ stellar black hole and a $10^9 M_{\odot}$ supermassive black hole? (Make sure your readers can clearly tell which is which.) [4 pts]
- (d) At what wavelengths would you expect to find the peaks of their spectra? And what are the names of those parts of the EM spectrum (e.g. Table 5.1)? [2 pts]

12.3 Stellar close encounters

Your text works through a calculation of an average time between collisions for stars in the solar neighborhood, finding that this time t_* is roughly 5×10^{10} Gyr. Notice that this calculation assumes the stars in question must get close enough to physically contact each other, and their

separation will be just twice their radii. But stars could seriously disrupt each others' planetary systems even if they don't get quite that close.

- (a) A reasonable estimate for the separation at which stars can disrupt each others' orbits and planetary systems is an effective gravitational scattering radius R_{eff} , given by $R_{\text{eff}} \sim 6Gm_*/v^2$. Here m_* is the mass of a star and v is a typical relative velocity between stars. For $v = 30 \text{ km s}^{-1}$, as in the text, and a star of mass $1 M_\odot$, how large is this effective gravitational radius? How does it compare to the size of the solar system, for example? [5 pts]
- (b) Use this effective radius rather than the physical radius to make a new estimate of the typical time between "collisions" of stars in the solar neighborhood. Rather than a physical collision, of course, it'll be the typical time that elapses before a star's orbit and/or its planetary system is disrupted by a nearby star. Assume the typical density of stars in the solar neighborhood is $n_* = 0.1 \text{ pc}^{-3}$. [5 pts]
- (c) If a typical star in the solar neighborhood is 5 Gyr old, what is the probability that this typical star has had its planetary system disrupted by one of these "collisions"? [3 pts]

12.4 Expansion of a Newtonian universe

Consider a model of a Newtonian universe in which the density matches the critical density $\rho_{c,0}$. In this case, the scale factor $a(t)$ is implicitly given by the relation

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G \rho_{c,0}}{3} \frac{1}{a^3}.$$

- (a) If $a(0) = 0$ and $a(t_0) = 1$, find an expression for $a(t)$. [5 pts]
- (b) What is t_0 in terms of the present-day Hubble constant H_0 ? [3 pts]
- (c) We observe a Hubble constant of $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and the oldest stars in the universe have an age of $t_* \approx 13 \text{ Gyr}$. Are those two observations consistent with a Newtonian universe that has $\rho_0 = \rho_{c,0}$? [5 pts]

EXTRA CREDIT

12.5 Olbers's paradox again

Imagine a universe full of cold, dead white dwarf cinders with masses around $0.7 M_\odot$ and radii around $0.01 R_\odot$.

- (a) What would be the number density of such WD cinders n_{WD} that would be required in order to make the mass density of this universe equal to the currently observed critical density, $\rho_{c,0} = 3H_0^2/(8\pi G)$? [2 pts]
- (b) Given this density n_{WD} , what would be the average distance to which you could see before your line of sight intersected a dead WD? [3 pts]
- (c) We can actually observe galaxies out to a distance of roughly 4000 Mpc, so evidently the universe is transparent on that length scale. Does this transparency place useful limits on the number density of WD cinders? [2 pts]
- (d) Does your answer depend on the density of the object with which the universe is filled? For example, if the universe was filled with basketballs, of a typical mass 0.6 kg and radius 0.12 m, how would your answers to the previous parts change? [3 pts]